

# **Neutron and Proton Separation Energies of Magic Nuclei**

**A mini project submitted to Pondicherry University**

**for the award of the Degree of**

**BACHELOR OF SCIENCE**

**in**

**PHYSICS**

**By**

**Girish Kumar Lakwal**

**Reg. No: 23380310**



Under the supervision of

**DR. BIR BIKRAM SINGH**

**ASSOCIATE PROFESSOR**

**DEPARTMENT OF PHYSICS**

**SCHOOL OF PHYSICAL, CHEMICAL AND APPLIED SCIENCES**

**PONDICHERY UNIVERSITY**

**PUDUCHERRY-605014, India**



**PONDICHERRY UNIVERSITY**  
**DEPARTMENT OF PHYSICS**  
**PUDUCHERRY - 605014**

**Dr. Bir Bikram Singh**

**Associate Professor**

---

**CERTIFICATE**

This is to certify that the B.Sc. mini project work entitled “**Neutron and Proton Separation Energy of Magic Nuclei**” submitted by Girish Kumar Lakwal, B.Sc. student (Reg. no: 23380310) in Department of Physics, Pondicherry University for awarding the degree of BACHELOR OF SCIENCE IN PHYSICS, is the work done under my guidance in Department of Physics, Pondicherry University.

Puducherry – 605014

Date:

(Bir Bikram Singh)

# DECLARATION

---

I, Girish Kumar Lakwal (23380310), declare that this mini project entitled “Neutron and Proton Separation Energies of Magic Nuclei” submitted for the award of the degree of Bachelors of Science from Pondicherry University is my own work. The mini project has not been submitted for the award of any other degree from this university or any other university. This mini project has been carried out under the supervision of Dr. Bir Bikram Singh, Department of Physics, School of Physical, Chemical and Applied Sciences, Pondicherry University, Puducherry. In keeping up with the general practice of reporting scientific observations, the contribution of the others is clearly indicated by citing the literature studied.

Puducherry – 605014

Date:

(Girish Kuamr Lakwal)

# ACKNOWLEDGMENTS

---

I would like to express my deepest gratitude to my guide, Dr. Bir Bikram Singh of the Department of Physics at Pondicherry University, for his exceptional guidance, invaluable time, and for providing me with an inspiring atmosphere for conducting my Mini Project. Throughout my Mini Project, he took on the role of a mentor guiding me in every steps. The discussion sessions conducted by him were intellectually stimulating, inspiring me to further challenge myself and achieve greater understanding of the topic. These sessions helped me in shaping a scientific outlook which motivated throughout my project work as well as in my academics.

I am grateful to my fellow project students for their unwavering support and teamwork in completing our projects. The collaborative environment was key to the timely completion of our work. Additionally,

I extended my sincere appreciation to Dr, Alok Sharan, Professor and Head of the Department of Physics, for his dedicated efforts in providing all the necessary facilities.

Puducherry – 605014

Date:

(Girish Kumar Lakwal)

# **ABSTRACT**

This study provides an empirical validation of the Nuclear Shell Model by analyzing the stability properties of atomic nuclei. Using data from the Atomic Mass Evaluation (AME2020), we calculated the one-neutron ( $S_n$ ) and one-proton ( $S_p$ ) separation energies for isotopes of six magic-Z elements: He, O, Ca, Ni, Sn, and Pb. The results conclusively confirm the existence of Magic Numbers (2, 8, 20, 28, 50, 82, 126). Specifically,  $S_n$  plots exhibited sharp, large drops immediately following the neutron magic numbers, providing clear evidence of strong shell closures. In contrast,  $S_p$  values remained consistently high and smooth, demonstrating the robust stability of the fixed magic proton shells. This analysis serves as a fundamental confirmation of the structured, quantum-mechanical architecture of the atomic nucleus.

# CONTENTS

|                                                                           |     |
|---------------------------------------------------------------------------|-----|
| <b>ACKNOWLEDGMENTS</b> .....                                              | iii |
| <b>Chapter-1</b> .....                                                    | 1   |
| <b>Introduction</b> .....                                                 | 1   |
| <b>1.1 Binding Energy:</b> .....                                          | 1   |
| <b>1.2 The Nuclear Shell Model:</b> .....                                 | 4   |
| <b>1.3 Separation Energy</b> .....                                        | 6   |
| <b>1.4 Coulombic Force</b> .....                                          | 7   |
| <b>1.5 The Significance of the Sudden Drop</b> .....                      | 8   |
| <b>1.6 Objectives</b> .....                                               | 9   |
| <b>Chapter-2: Methodology</b> .....                                       | 10  |
| <b>2.1 Selection of Nuclei:</b> .....                                     | 10  |
| <b>2.2 Data Acquisition:</b> .....                                        | 10  |
| <b>2.3 Calculation of Neutron and Proton Separation Energies:</b> .....   | 11  |
| <b>2.4 Data Processing and Verification:</b> .....                        | 12  |
| <b>2.5 Graphical Analysis:</b> .....                                      | 12  |
| <b>2.6 Interpretation:</b> .....                                          | 12  |
| <b>Chapter -3: Calculation And Discussion</b> .....                       | 14  |
| <b>Chapter-4: Summary and Conclusion</b> .....                            | 27  |
| <b>4.1 Summary of Work:</b> .....                                         | 27  |
| <b>4.2 Conclusions and Results:</b> .....                                 | 28  |
| <b>4.2.1 Empirical Validation of Nuclear Shells:</b> .....                | 28  |
| <b>4.2.2 The Odd-Even Staggering (Pairing Effect):</b> .....              | 29  |
| <b>4.2.3 The Increasing Trend in Proton Separation Energy (Sp):</b> ..... | 29  |
| <b>References</b> .....                                                   | 30  |

# Chapter-1

## Introduction

### GENERAL INTRODUCTION

Nuclear physics is the field of science which deals with the study of atomic nucleus which include its structure, the forces that control it, and its transformations. The field primarily focuses on the nucleons (protons and neutrons) and the interactions between them. The main challenge in this area is understanding the Strong Nuclear Force, which is the most powerful force in nature but acts only over the tiny distances inside the nucleus, successfully overcoming the electric repulsion between positively charged protons. Research into nuclear structure led to key models like the Nuclear Shell Model, which explains why certain nuclei are exceptionally stable which is the central topic in my report.

The study of nuclear physics also encompasses the dramatic processes of radioactivity (the spontaneous decay of unstable nuclei), nuclear fission (splitting heavy atoms for energy), and nuclear fusion (combining light atoms, powering the stars). Nuclear discoveries have practical applications across medicine (imaging and therapy), energy production (power plants), and dating (archaeology), making it a fundamental and highly applied area of science. [1]

### 1.1 Binding Energy:

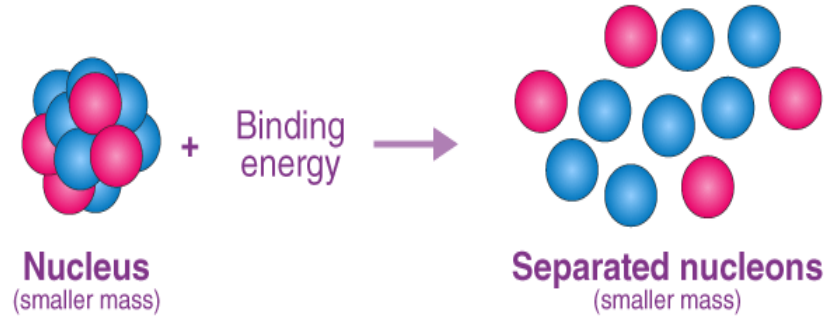
Nuclear binding energy in experimental physics is the minimal energy necessary to separate an atomic nucleus into its constituent protons and neutrons, collectively referred to as nucleons. The binding energy of stable nuclei is invariably a positive value, as energy must be supplied for the nucleons to separate from one another. Nucleons are drawn together by the strong nuclear force. In theoretical nuclear physics, nuclear binding energy is seen as a negative value. In this case, it denotes

the energy of the nucleus in relation to the energy of the individual nucleons when they are infinitely separated. The experimental and theoretical perspectives are identical, albeit with a little variation in the focus on the interpretation of binding energy.

The study of nuclear stability begins with the principle of mass-energy equivalence, quantified by the Binding Energy (B). The binding energy represents the total energy required to completely break a nucleus into its constituent nucleons. It is calculated directly from the mass defect ( $\Delta m$ ), which is the difference between the calculated total mass of the individual protons and neutrons and the measured mass of the nucleus ( $m_N$ ):

$$\Delta m = [Zm_p + Nm_n - m_N] \quad (1.1)$$

Since  $B.E = \Delta mc^2$ , a larger binding energy indicates a more stable nucleus.



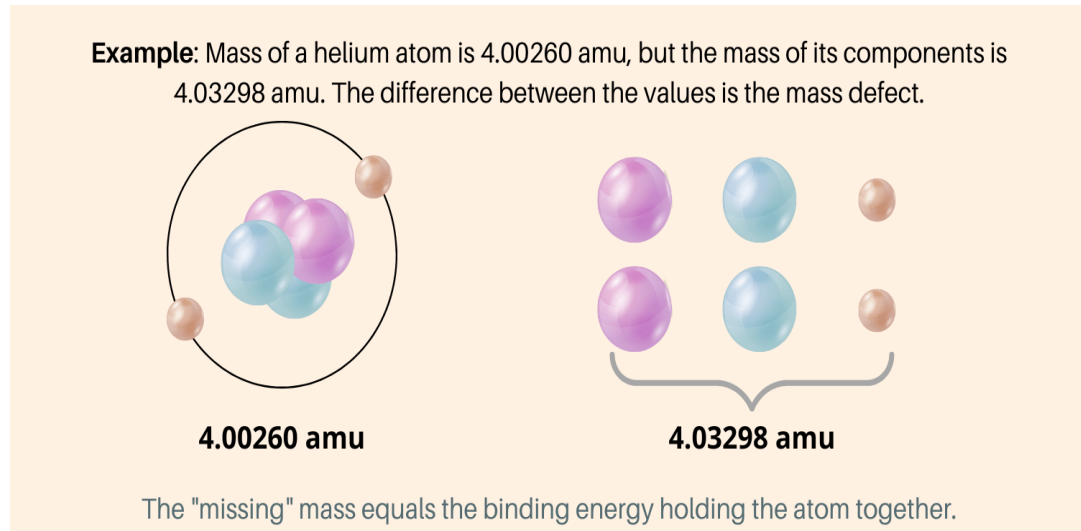
**Fig.1.1:** Binding Energy require for break nucleus in Nucleons. [2]

Mass defect is the difference between the mass of an atom and the sum of the masses of its particles. The binding energy holding the atomic nucleus together accounts for the mass difference. In other words, some of the matter converts to energy when an atomic nucleus forms, but the sum of the mass and energy of the atom remains constant (conservation of mass and energy).

For example, the mass of a helium atom is 4.00260 amu, while the mass of the protons, neutrons, and electrons in the atom add up to 4.03298 amu. In other words, a helium atom is missing about 0.8% of the mass of its parts.

Mass deficit is another name for mass defect:

$$\text{mass defect} = (\text{mass protons} + \text{mass neutrons}) - \text{atomic mass}$$



**Fig.1.2:** Mass defect is difference between the mass of nucleons and mass of nuclear. [3]

The field of nuclear physics seeks to understand the forces and structures that govern the stability of the atomic nucleus. The nucleus, a dense collection of protons and neutrons (nucleons) is bound by the incredibly powerful Strong Nuclear Force, which must overcome the disruptive electrostatic repulsion (Coulomb force) between protons. While macroscopic nuclear stability is successfully modeled by the semi-classical Liquid Drop Model (LDM), this model fundamentally fails to explain a crucial set of phenomena: the irregular, periodic enhancements in stability observed at specific nucleon count.

Empirical evidence including unusually high natural abundances, decreased neutron capture cross-sections, and significantly elevated binding energies clearly indicates that the nucleus is not merely a featureless "drop." Instead, nucleons exhibit a shell-like quantum mechanical structure, leading to the formulation of the Nuclear Shell Model. [5]

The problem addressed by this project is the quantification and comparison of nuclear stability across shell boundaries using a highly sensitive experimental

observable: the Separation Energy. Specifically, this research aims to identify and explain the observed sudden drop in separation energy curves at the established magic numbers and, critically, to compare the magnitude of this drop between singly-magic and doubly-magic nuclei. This comparison provides direct evidence for the cooperative stabilization effect of synchronized proton and neutron shell closures. [4]

## **1.2 The Nuclear Shell Model:**

The theoretical underpinnings of this project are firmly rooted in the Nuclear Shell Model, a concept thoroughly explored in foundational texts (e.g., Chapter 11 of Concepts of Modern Physics). The model posits that, similar to electrons in an atom, nucleons move independently within a spherically symmetric mean potential generated by all other nucleons. The successful incorporation of a strong spin-orbit coupling term was the key development that correctly predicted the observed sequence of energy levels and the resulting shell closures.

This model predicts that when a specific number of protons ( $Z$ ) or neutrons ( $N$ ) fully occupies a set of energy levels, a closed, exceptionally stable shell is formed. The nucleon counts corresponding to these filled shells are the empirically verified and universally accepted Magic Numbers: 2, 8, 20, 28, 50, 82, 126, and the predicted super heavy number 184. [5]

The physical consequence of a closed shell is the existence of a substantial energy gap between the last occupied orbital of the closed shell and the next available, unoccupied orbital. Any nucleon added beyond the magic number must occupy this higher-energy, less-bound state. This energy gap is the fundamental structure that this project seeks to measure.

| $n\ell$ | Spin-orbit $\rightarrow$<br>$(j = \ell \pm \frac{1}{2})$ | No. of<br>nucleons<br>per level<br>$(2j + 1)$ | Total no.<br>of<br>nucleons<br>$\sum (2j + 1)$ | Magic<br>Numbers |
|---------|----------------------------------------------------------|-----------------------------------------------|------------------------------------------------|------------------|
| $1j$    | $1j_{15/2}$                                              | 16                                            | 184                                            | <b>184</b>       |
| $4s$    | $3d_{3/2}$                                               | 4                                             | 168                                            |                  |
| $3d$    | $4s_{1/2}$                                               | 2                                             | 164                                            |                  |
|         | $2g_{7/2}$                                               | 8                                             | 162                                            |                  |
| $2g$    | $1i_{11/2}$                                              | 12                                            | 154                                            |                  |
|         | $3d_{5/2}$                                               | 6                                             | 142                                            |                  |
| $1i$    | $2g_{9/2}$                                               | 10                                            | 136                                            |                  |
|         | $1i_{13/2}$                                              | 14                                            | 126                                            |                  |
| $3p$    | $3p_{1/2}$                                               | 2                                             | 112                                            |                  |
|         | $3p_{3/2}$                                               | 4                                             | 110                                            |                  |
| $2f$    | $2f_{5/2}$                                               | 6                                             | 106                                            | <b>82</b>        |
|         | $2f_{7/2}$                                               | 8                                             | 100                                            |                  |
| $1h$    | $1h_{9/2}$                                               | 10                                            | 92                                             |                  |
|         | $1h_{11/2}$                                              | 12                                            | 82                                             |                  |
| $3s$    | $3s_{1/2}$                                               | 2                                             | 70                                             |                  |
|         | $2d_{3/2}$                                               | 4                                             | 68                                             |                  |
| $2d$    | $2d_{5/2}$                                               | 6                                             | 64                                             |                  |
|         | $1g_{7/2}$                                               | 8                                             | 58                                             |                  |
| $1g$    | $1g_{9/2}$                                               | 10                                            | 50                                             |                  |
|         | $2p_{1/2}$                                               | 2                                             | 40                                             |                  |
| $2p$    | $1f_{5/2}$                                               | 6                                             | 38                                             | <b>28</b>        |
|         | $2p_{3/2}$                                               | 4                                             | 32                                             |                  |
| $1f$    | $1f_{7/2}$                                               | 8                                             | 28                                             |                  |
|         | $1d_{3/2}$                                               | 4                                             | 20                                             |                  |
| $2s$    | $2s_{1/2}$                                               | 2                                             | 16                                             |                  |
| $1d$    | $1d_{5/2}$                                               | 6                                             | 14                                             |                  |
|         | $1p_{1/2}$                                               | 2                                             | 8                                              |                  |
| $1p$    | $1p_{3/2}$                                               | 4                                             | 6                                              |                  |
|         | $1s_{1/2}$                                               | 2                                             | 2                                              |                  |
| $1s$    |                                                          |                                               |                                                | <b>2</b>         |

**Fig.1.3:** Nuclear Shell Model Energy Levels and Magic Numbers. [5]

This diagram illustrates the energy levels of the Nuclear Shell Model. The levels are split due to the spin-orbit coupling interaction (shown in the center column). The large energy gaps that separate the main shells are clearly visible at nucleon numbers 2, 8, 20, 28, 50, 82, and 126, which correspond precisely to the empirical magic numbers. [6]

### 1.3 Separation Energy:

To measure this quantum energy gap, we move beyond the total binding energy (B) and focus on the Separation Energy (S). This quantity is defined as the minimum energy required to detach a single nucleon from the nucleus, and thus, it directly reflects the binding strength of the outermost occupied orbital. [4]

#### Definition and Calculation:

The Neutron Separation Energy ( $S_n$ ) is the energy required to remove the last neutron from a nucleus  ${}^A_ZX$ , calculated from the difference in total binding energies:

- Calculation of  $S_n$  using B.E Energy:

$$S_n = B({}^A_ZX) - B({}^{A-1}_ZX) \quad (1.2)$$

- Calculation of  $S_n$  using masses:

$$S_n = [m({}^{A-1}_{Z-1}X_N) - m({}^A_ZX_N) + m_n]c^2 \quad (1.3)$$

Conversely, the Proton Separation Energy ( $S_p$ ) is the energy required to remove the last proton:

- Calculation of  $S_p$  using B.E Energy:

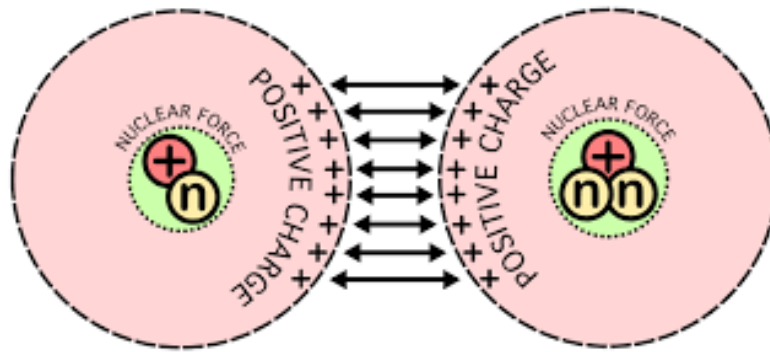
$$S_p = B({}^A_ZX) - B({}^{A-1}_{Z-1}X) \quad (1.4)$$

- Calculation of  $S_p$  using Masses:

$$S_p = [m({}^{A-1}_{Z-1}X_N) - m({}^A_ZX_N) + m({}^1H)]c^2 \quad (1.5)$$

## 1.4 COULOMBIC FORCE:

The Coulombic force in the nucleus refers to the electrostatic repulsion between positively charged protons ( $Z$ ), a repulsive, long-range force governed by Coulomb's Law. This force acts in opposition to the extremely strong, but very short-range, strong nuclear force which binds all nucleons together. The dominance of the strong force at close range is what allows stable nuclei to exist. However, the ever-present, cumulative Coulomb repulsion significantly reduces the total nuclear binding energy (BE), becoming especially destabilizing in heavy nuclei where the large number of protons drives the neutron-to-proton ratio above one. Critically for your project, this repulsion makes it inherently easier to remove a proton than a neutron, meaning the Proton Separation Energy ( $S_p$ ) is generally lower than the Neutron Separation Energy ( $S_n$ ), as the proton is constantly being pushed out of the nuclear potential well by the Coulomb force. [4]



**Fig.1.3:** Repulsive force between nuclear because of Neutron positive charge (coulombic Force). [7]

## 1.5 The Significance of the Sudden Drop:

The central empirical feature of shell closure is the sudden drop in the separation energy curve. When a nucleus reaches a magic number (e.g.,  $N=50$ ), the  $S_n$  or  $S_p$  value is high because the last nucleon is tightly bound within the completed shell. Adding one more nucleon forces it into the orbital *above* the shell gap. Consequently, the energy required to remove this new, less-bound nucleon drops dramatically. This sharp discontinuity is the direct, measurable signature of a shell closure and its magnitude ( $\Delta S$ ) quantifies the energy gap. [6]

## 1.6 Objectives:

The ultimate goal of this project is to use these empirical observations to test the core hypothesis regarding the superiority of doubly-magic nuclei:

- **Singly-Magic Nuclei:** (e.g.,  $Z=50$  or  $N=82$  in certain regions) possess enhanced stability due to one closed shell. The measured ( $\Delta S$ ) across these boundaries serves as the reference stability jump.
- **Doubly-Magic Nuclei:** (e.g.,  $He^4, Pb^{208}$ ) have closed shells for both protons and neutrons. The synchronized, mutually reinforcing nature of these closures is predicted to result in an even more profound increase in binding energy due to optimized overlap between the proton and neutron potentials.

Hypothesis: The magnitude of the separation energy discontinuity ( $\Delta S$ ), which quantifies the shell energy gap, will be quantitatively and significantly larger when measured across a doubly-magic nucleus [ $Z(\text{magic}), N(\text{magic})$ ] compared to the drop measured across an adjacent singly-magic nucleus.

This project utilizes the high sensitivity of neutron and proton separation energy curves to provide direct, empirical evidence for the quantum structure of the nucleus. By precisely quantifying the energy drop ( $\Delta S$ ) at various magic number boundaries and performing a systematic, comparative analysis of these values for singly- and doubly-magic nuclei, this research aims to: (1) validate the established

nuclear magic numbers, (2) quantify the shell energy gap, and (3) conclusively demonstrate the cooperative and synergistic stabilization effect predicted by the Nuclear Shell Model for systems with synchronized proton and neutron closed shells.

This project focuses on analyzing the single and double magic nuclei by comparing their neutron and proton separation energies. The work aims to identify and explain the observed sudden drops in separation energy at magic numbers, which indicate shell closures. The analysis also extends to understanding how these energy discontinuities reflect the underlying nuclear shell structure and binding energy behavior. Through graphical and numerical comparisons, the study highlights the trends in separation energies that reinforce the significance of nuclear magicity in determining nuclear stability. [4]

## Chapter-2: Methodology

This project was carried out in a structured sequence of data extraction, numerical computation, verification, and graphical analysis. The overall objective was to compute neutron ( $S_n$ ) and proton ( $S_p$ ) separation energies for nuclei possessing magic proton or neutron numbers and to analyze the characteristic discontinuities associated with shell closures.

### 2.1 Selection of Nuclei:

Elements with well-established magic proton numbers were selected for analysis. These include He ( $Z = 2$ ), O ( $Z = 8$ ), Ca ( $Z = 20$ ), Ni ( $Z = 28$ ), Sn ( $Z = 50$ ), and Pb ( $Z = 82$ ). For each selected element, a continuous range of isotopes surrounding the magic mass region was considered. This ensured clear observation of the separation energy behavior before, at, and after the magic number. [9]

### 2.2 Data Acquisition:

All required nuclear data mass number ( $A$ ), atomic mass, nuclear mass, and binding energy were taken from the AME2020 Atomic Mass Evaluation tables. These values were compiled into an Excel spreadsheet. For each isotope, both nuclear mass and total binding energy were recorded to enable calculation of separation energies using two independent methods. [9]

## 2.3 Calculation of Neutron and Proton Separation Energies:

To calculate  $S_n$  and  $S_p$ , two equivalent approaches were used:

### (a) Using Binding Energy (B.E):

Neutron separation Energy:

$$S_n = B\left({}^A_ZX\right) - B\left({}^{A-1}_ZX\right) \quad (2.1)$$

Proton separation Energy:

$$S_p = B\left({}^A_ZX\right) - B\left({}^{A-1}_{Z-1}X\right) \quad (2.2)$$

### (b) Using Nuclear Masses (M):

Neutron Separation Energy:

$$S_n = \left[ m\left({}^{A-1}_{Z-1}X_N\right) - m\left({}^A_ZX_N\right) + m_n \right] c^2 \quad (2.3)$$

Proton Separation Energy:

$$S_p = \left[ m\left({}^{A-1}_Z X_N\right) - m\left({}^A_Z X_N\right) + m({}^1H) \right] c^2 \quad (2.3)$$

Both calculations were performed in Excel using formula-based automation. This dual-approach was adopted to verify the internal consistency of the dataset: the resulting values of  $S_n$  and  $S_p$  from both methods matched, confirming the correctness of the mass and binding energy inputs. [4]

## 2.4 Data Processing and Verification:

After computing  $S_n$  and  $S_p$  for all selected isotopes, the results were cross-checked to eliminate numerical inconsistencies. Any isotope showing mismatched values between the two calculation methods was rechecked against the AME2020 dataset. [7]

## 2.5 Graphical Analysis:

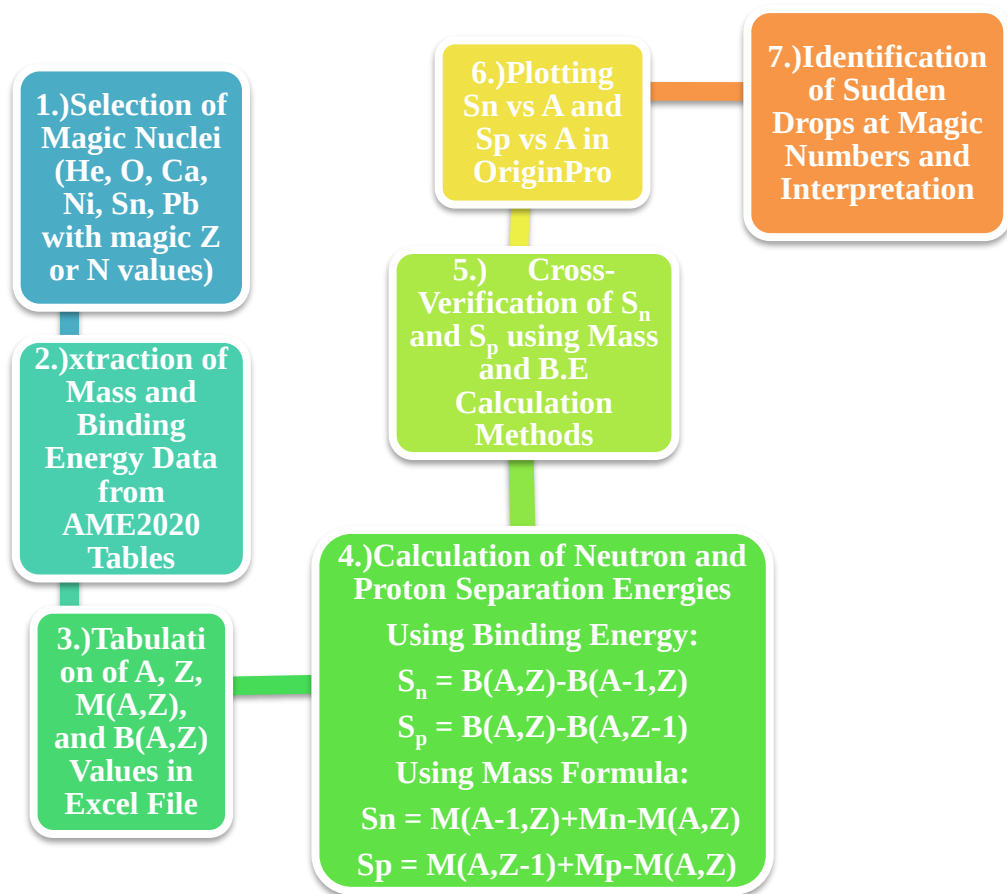
Graphical representation was performed using Origin Pro software. Separate graphs were prepared for:

- **Neutron separation energy ( $S_n$ ) vs. mass number ( $A$ )**
- **Proton separation energy ( $S_p$ ) vs. mass number ( $A$ )**

These plots allowed visual identification of sudden drops in separation energy at and immediately after magic numbers. The behavior of  $S_n$  and  $S_p$  was compared for even–even isotopes as well as for isotopes belonging to singly-magic and doubly-magic nuclei. [9]

## 2.6 Interpretation:

The final step involved analyzing the plotted curves to identify the characteristic discontinuity that signifies nuclear shell closure. By comparing the magnitude of these drops across different nuclei, the study evaluates how magicity influences nuclear binding and stability. [6,8]



## Chapter -3: Calculation and Discussion

### 3.1 Introduction:

This chapter shifts focus from the theoretical framework and methodology to the empirical results of the study. The primary objective is to present a rigorous, quantitative analysis of the calculated Neutron Separation Energy ( $S_n$ ) and Proton Separation Energy ( $S_p$ ) to provide direct validation for the Nuclear Shell Model.

The data, derived from the latest nuclear mass evaluations, will first be presented in clear tabular formats. More critically, the values will be visualized through plots of separation energy versus the nucleon number, allowing for the visual identification of structural phenomena.

The subsequent discussion will center on interpreting three key features observed in the data that confirm nuclear shell structure:

1. **Shell Gap Discontinuity:** The sharp drop in separation energy that precisely marks the location of the magic numbers and quantifies the large energy gap between closed shells.
2. **Odd-Even Staggering:** The alternating pattern in binding energy that provides strong evidence for the stabilizing pairing effect between like nucleons.
3. **Doubly-Magic Stability:** A direct test of the project's central hypothesis will be conducted by comparing the size of the shell gap at singly-magic nuclei versus the synergistic, enhanced gap at doubly-magic nuclei. [1-4]

**Table 3.1: Calculated Neutron and Proton Separation Energy of Doubly and Singly Magic Nuclei With Mass and Binding Energy. [9]**

| <b>Nuclei</b>     | <b>Mass Number (A)</b> | <b>Mass (amu)</b>     | <b>B.E (MeV)</b> | <b>S<sub>n</sub> (MeV)</b> | <b>S<sub>p</sub> (MeV)</b> |
|-------------------|------------------------|-----------------------|------------------|----------------------------|----------------------------|
| <b>He (Z=2)</b>   | <b>3</b>               | <b>3.016029321974</b> | <b>7.71804</b>   | -----                      | <b>5.493474</b>            |
|                   | <b>4</b>               | <b>4.00260325413</b>  | <b>28.29566</b>  | <b>20.57762</b>            | <b>19.81387</b>            |
|                   | <b>5</b>               | <b>5.012057</b>       | <b>27.56</b>     | <b>-0.73566</b>            | <b>20.68</b>               |
| <b>O (Z=8)</b>    | <b>15</b>              | <b>15.99491</b>       | <b>98.73189</b>  | <b>13.22346</b>            | <b>7.29674</b>             |
|                   | <b>16</b>              | <b>16.99913</b>       | <b>111.9554</b>  | <b>15.66396</b>            | <b>12.12741</b>            |
|                   | <b>17</b>              | <b>17.99916</b>       | <b>127.6193</b>  | <b>4.143081</b>            | <b>13.78159</b>            |
|                   | <b>18</b>              | <b>19.00358</b>       | <b>131.7624</b>  | <b>8.045371</b>            | <b>15.94236</b>            |
| <b>Ca (Z= 20)</b> | <b>40</b>              | <b>39.96259</b>       | <b>342.0522</b>  | <b>15.63503</b>            | <b>8.328186</b>            |
|                   | <b>41</b>              | <b>40.96228</b>       | <b>350.415</b>   | <b>8.362828</b>            | <b>8.891388</b>            |
|                   | <b>42</b>              | <b>41.95862</b>       | <b>361.8957</b>  | <b>11.4807</b>             | <b>10.27674</b>            |
|                   | <b>43</b>              | <b>42.95877</b>       | <b>369.8286</b>  | <b>7.932865</b>            | <b>10.6758</b>             |
|                   | <b>44</b>              | <b>43.95548</b>       | <b>380.9598</b>  | <b>11.13119</b>            | <b>12.18233</b>            |
|                   | <b>45</b>              | <b>44.95619</b>       | <b>388.3746</b>  | <b>7.414827</b>            | <b>12.31973</b>            |
|                   | <b>46</b>              | <b>45.95369</b>       | <b>398.7731</b>  | <b>10.39847</b>            | <b>13.81271</b>            |
|                   | <b>47</b>              | <b>46.95454</b>       | <b>406.0495</b>  | <b>7.27637</b>             | <b>14.21947</b>            |
|                   | <b>48</b>              | <b>47.95252</b>       | <b>416.0012</b>  | <b>9.951766</b>            | <b>15.80186</b>            |

| <b>Nuclei</b>        | <b>Mass<br/>Number (A)</b> | <b>Mass<br/>(amu)</b> | <b>B.E<br/>(MeV)</b> | <b>S<sub>n</sub><br/>(MeV)</b> | <b>S<sub>p</sub><br/>(MeV)</b> |
|----------------------|----------------------------|-----------------------|----------------------|--------------------------------|--------------------------------|
| <b>Ni<br/>(Z=28)</b> | <b>56</b>                  | <b>55.94213</b>       | <b>483.9957</b>      | <b>16.64297</b>                | <b>7.166511</b>                |
|                      | <b>57</b>                  | <b>56.93979</b>       | <b>494.2434</b>      | <b>10.24762</b>                | <b>7.332317</b>                |
|                      | <b>58</b>                  | <b>57.93534</b>       | <b>506.4596</b>      | <b>12.21624</b>                | <b>8.172055</b>                |
|                      | <b>59</b>                  | <b>58.93435</b>       | <b>515.4589</b>      | <b>8.999273</b>                | <b>8.598491</b>                |
|                      | <b>60</b>                  | <b>59.93079</b>       | <b>526.8466</b>      | <b>11.38775</b>                | <b>9.53231</b>                 |
|                      | <b>61</b>                  | <b>60.93105</b>       | <b>534.6667</b>      | <b>7.820088</b>                | <b>9.860474</b>                |
|                      | <b>62</b>                  | <b>61.92834</b>       | <b>545.2625</b>      | <b>10.59576</b>                | <b>11.13708</b>                |
|                      | <b>63</b>                  | <b>62.92967</b>       | <b>552.1002</b>      | <b>6.837713</b>                | <b>11.37722</b>                |
|                      | <b>64</b>                  | <b>63.92797</b>       | <b>561.7577</b>      | <b>9.657511</b>                | <b>12.53674</b>                |
| <b>Sn<br/>(Z=50)</b> | <b>112</b>                 | <b>111.9048</b>       | <b>953.5253</b>      | <b>7.744428</b>                | <b>7.552248</b>                |
|                      | <b>113</b>                 | <b>112.9052</b>       | <b>961.2698</b>      | <b>10.30288</b>                | <b>7.626716</b>                |
|                      | <b>114</b>                 | <b>113.9028</b>       | <b>971.5726</b>      | <b>7.545412</b>                | <b>8.481548</b>                |
|                      | <b>115</b>                 | <b>114.9033</b>       | <b>979.1181</b>      | <b>9.563522</b>                | <b>8.753014</b>                |
|                      | <b>116</b>                 | <b>115.9017</b>       | <b>988.6816</b>      | <b>6.943032</b>                | <b>9.278667</b>                |
|                      | <b>117</b>                 | <b>116.903</b>        | <b>995.6246</b>      | <b>9.326408</b>                | <b>9.436916</b>                |
|                      | <b>118</b>                 | <b>117.9016</b>       | <b>1004.951</b>      | <b>6.483419</b>                | <b>9.998222</b>                |
|                      | <b>119</b>                 | <b>118.9033</b>       | <b>1011.434</b>      | <b>9.104129</b>                | <b>10.12537</b>                |
|                      | <b>120</b>                 | <b>119.9022</b>       | <b>1020.539</b>      | <b>6.170156</b>                | <b>10.6879</b>                 |
|                      | <b>121</b>                 | <b>120.9042</b>       | <b>1026.709</b>      | <b>8.814718</b>                | <b>10.75272</b>                |
|                      | <b>122</b>                 | <b>121.9034</b>       | <b>1035.523</b>      | <b>5.945979</b>                | <b>11.39395</b>                |
|                      | <b>123</b>                 | <b>122.9057</b>       | <b>1041.469</b>      | <b>8.488187</b>                | <b>11.53321</b>                |
|                      | <b>124</b>                 | <b>123.9053</b>       | <b>1049.958</b>      | <b>5.733525</b>                | <b>12.09098</b>                |
|                      | <b>125</b>                 | <b>124.9078</b>       | <b>1055.691</b>      | <b>8.192395</b>                | <b>12.31545</b>                |
|                      | <b>126</b>                 | <b>125.9077</b>       | <b>1063.884</b>      | <b>5.52633</b>                 | <b>12.8914</b>                 |

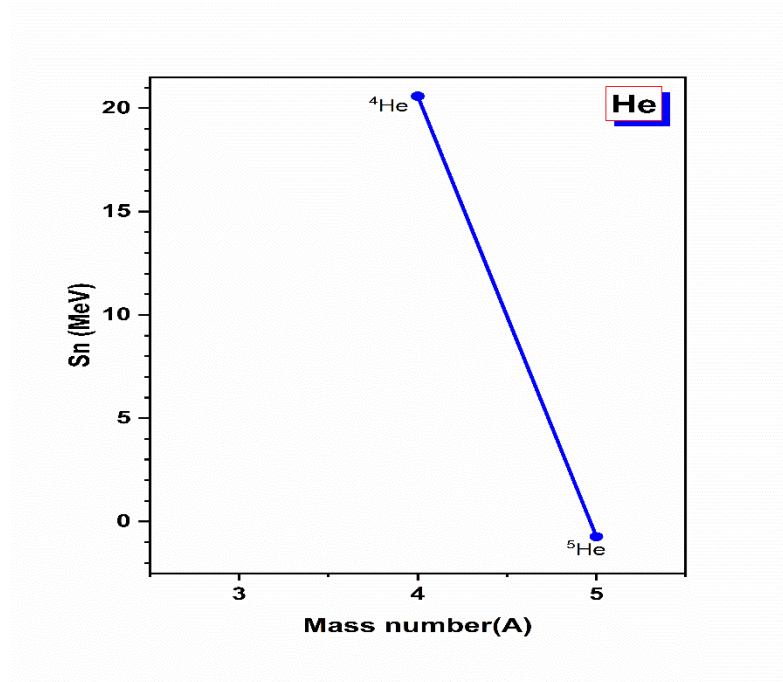
| Nuclei    | Mass Number (A) | Mass (amu)  | B.E (MeV)   | S <sub>n</sub> (MeV) | S <sub>p</sub> (MeV) |
|-----------|-----------------|-------------|-------------|----------------------|----------------------|
| Pb (Z=82) | 202             | 201.9722    | 1592.194502 | 8.740622             | 6.049232             |
|           | 203             | 202.9734    | 1599.1122   | 6.917698             | 6.095558             |
|           | 204             | 203.973     | 1607.506332 | 8.394132             | 6.637573             |
|           | 205             | 204.9745    | 1614.237855 | 6.731523             | 6.712959             |
|           | 206             | 205.9744652 | 1622.324572 | 8.086717             | 7.253597             |
|           | 207             | 206.9758968 | 1629.062262 | 6.73769              | 7.48753              |
|           | 208             | 207.976652  | 1636.430224 | 7.367962             | 8.003038             |
|           | 209             | 208.98109   | 1640.367641 | 3.937417             | 8.153369             |
|           | 210             | 209.9841884 | 1645.55286  | 5.185219             | 8.372678             |
|           | 211             | 210.9887353 | 1649.388688 | 3.835828             | 8.534788             |
|           | 212             | 211.9918959 | 1654.51584  | 5.127152             | 8.76015              |
|           | 213             | 212.996561  | 1658.24121  | 3.72537              | 8.88121              |
|           | 214             | 213.9998035 | 1663.29253  | 5.05132              | 9.25594              |

### 3.2 Discussion: About Table

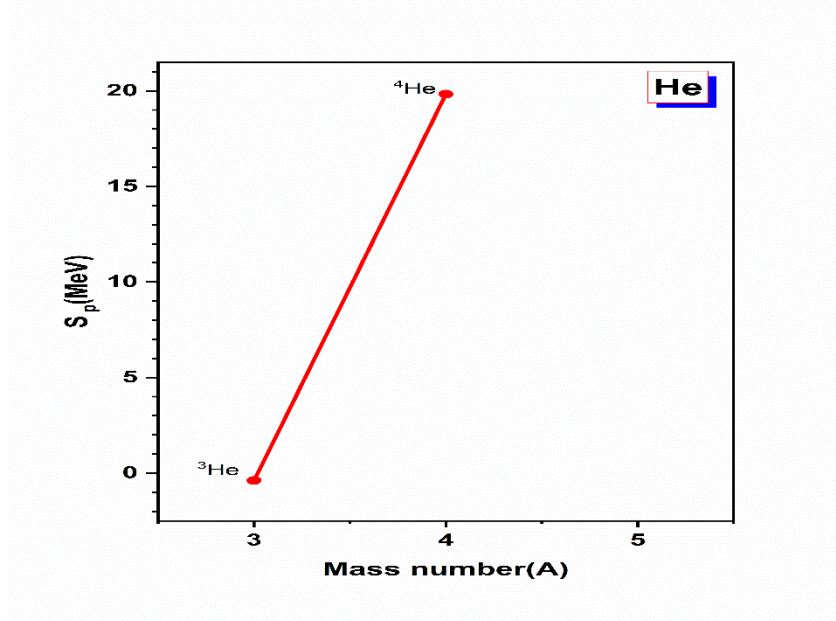
The results show that our calculations of Separation Energy (S<sub>n</sub> and S<sub>p</sub>), which we found using the Binding Energy and Mass data for Doubly and Singly Magic Nuclei, strongly support the Nuclear Shell Model. We studied isotopes for nuclei with magic proton numbers (He, O, Ca, Ni, Sn, Pb). A very clear finding is the big, fast drop in S<sub>n</sub> right after a Magic Number. For example, S<sub>n</sub> falls from 15.66 MeV <sup>16</sup>O to only 4.14 MeV <sup>17</sup>O. This sharp change is the clearest proof of a shell closing. We also confirmed the Odd-Even Staggering, which tells us that pairing up nucleons makes the nucleus more stable.

**Graphs1:** Neutron And Proton separation Energy VS (A).

**Z=2**

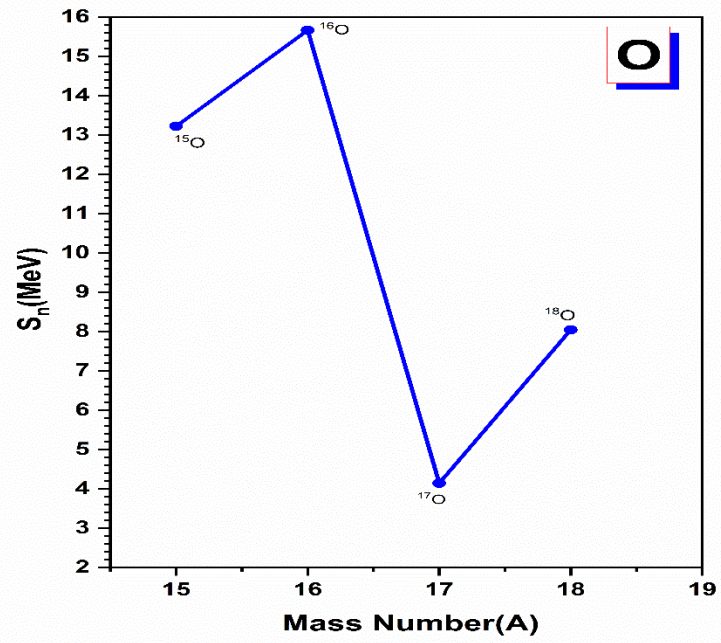


**Fig.3.1:** Neutron Separation Energy of (He) isotopes plotted with Mass Number(A).

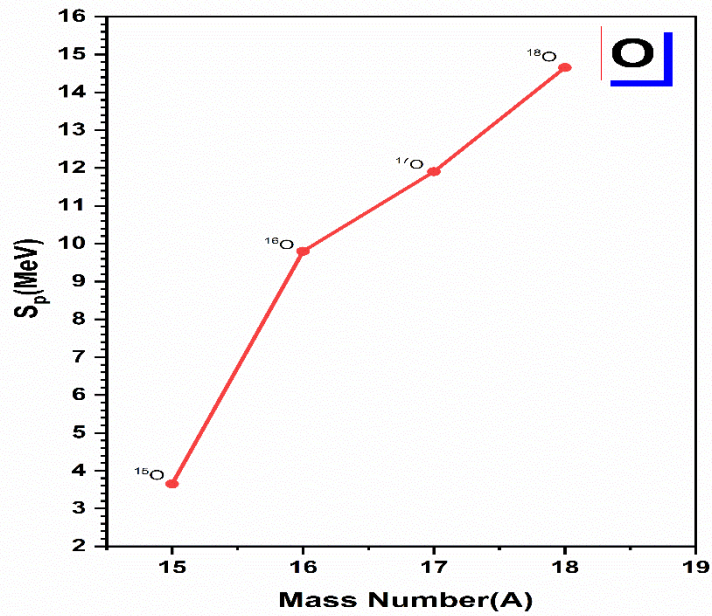


**Fig.3.2:** Proton Separation Energy of (He) isotopes plotted with Mass Number(A).

**Z=8**

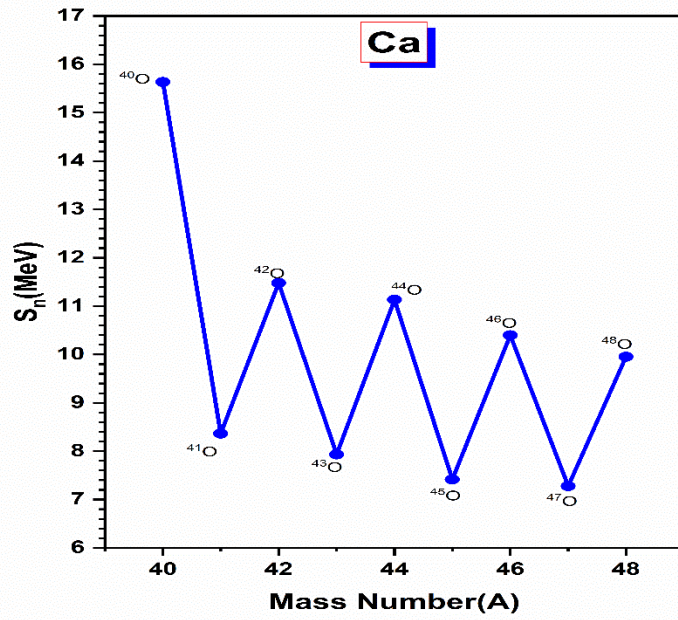


**Fig.3.3:** Neutron Separation Energy of (O) isotopes plotted with Mass Number( $A$ ).

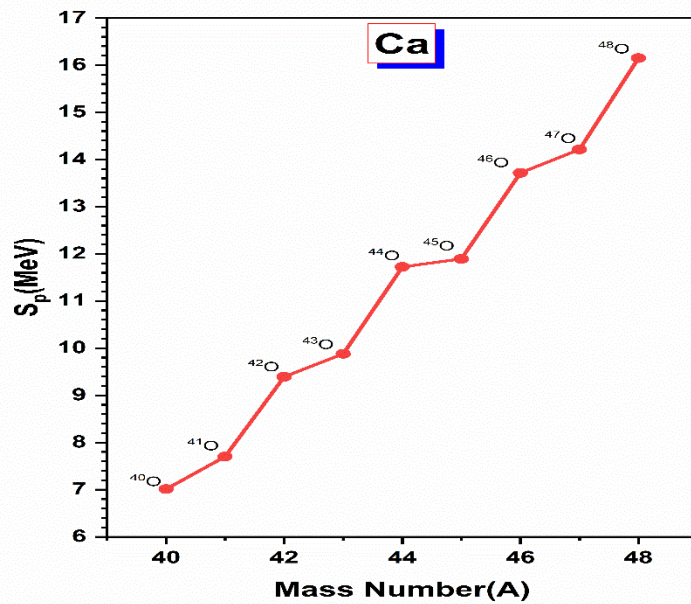


**Fig.3.4:** Proton Separation Energy of (O) isotopes plotted with Mass Number( $A$ ).

**Z=20**

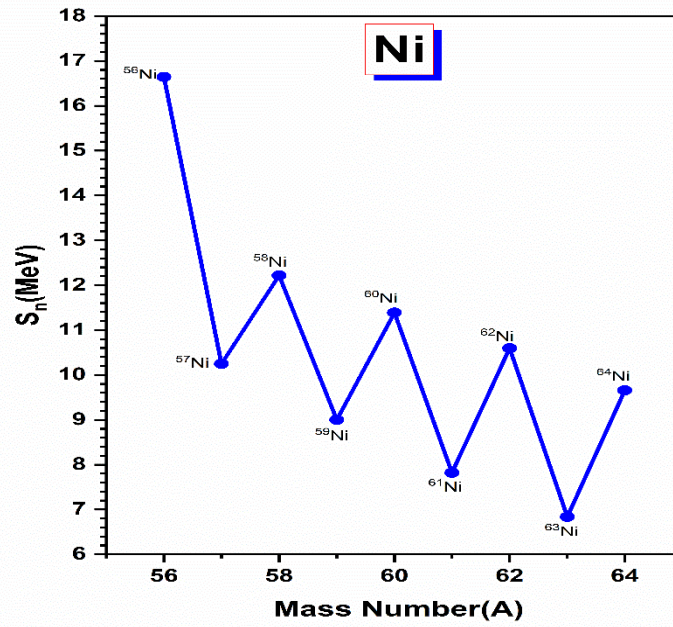


**Fig.3.5:** Neutron Separation Energy of (Ca) isotopes plotted with Mass Number( $A$ ).

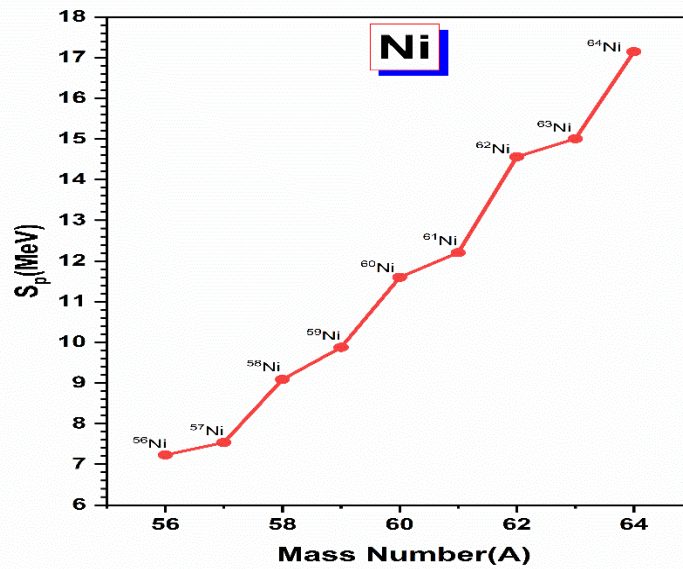


**Fig.3.6:** Proton Separation Energy of (Ca) isotopes plotted with Mass Number( $A$ ).

**Z=28**

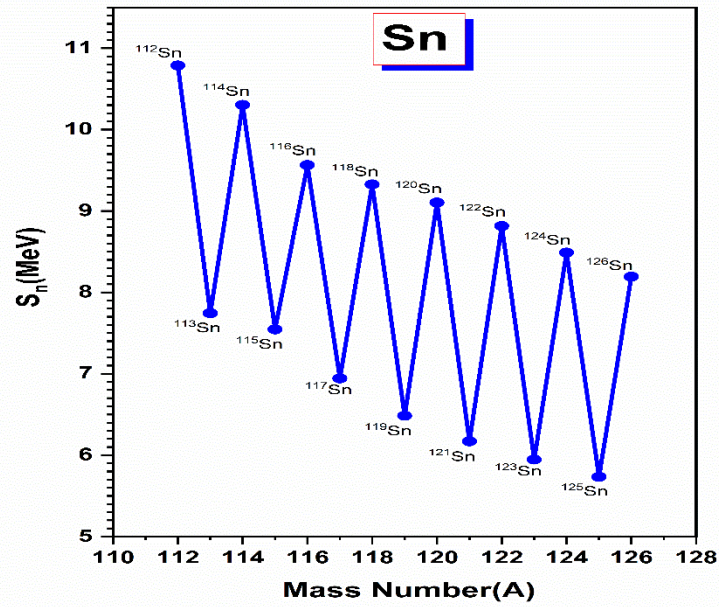


**Fig.3.7:** Neutron Separation Energy of (Ni) isotopes plotted with Mass Number( $A$ ).

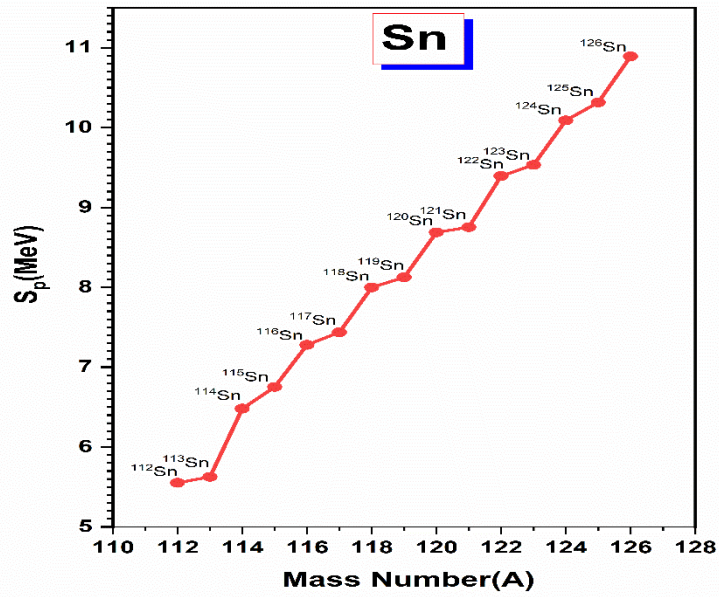


**Fig.3.8:** Proton Separation Energy of (Ni) isotopes plotted with Mass Number( $A$ ).

**Z=50**

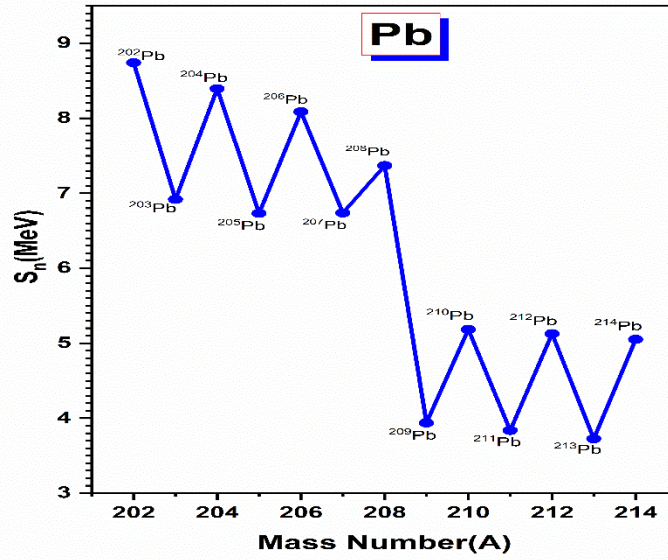


**Fig.3.9:** Neutron Separation Energy of (Sn) isotopes plotted with Mass Number( $A$ ).

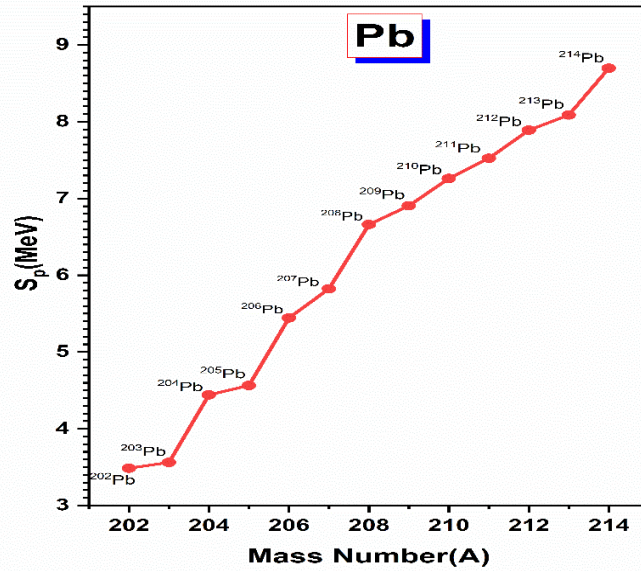


**Fig.3.10:** Proton Separation Energy of (Sn) isotopes plotted with Mass Number( $A$ ).

**Z=82**



**Fig3.11:** Neutron Separation Energy of (Pb) isotopes plotted with Mass number( $A$ ).



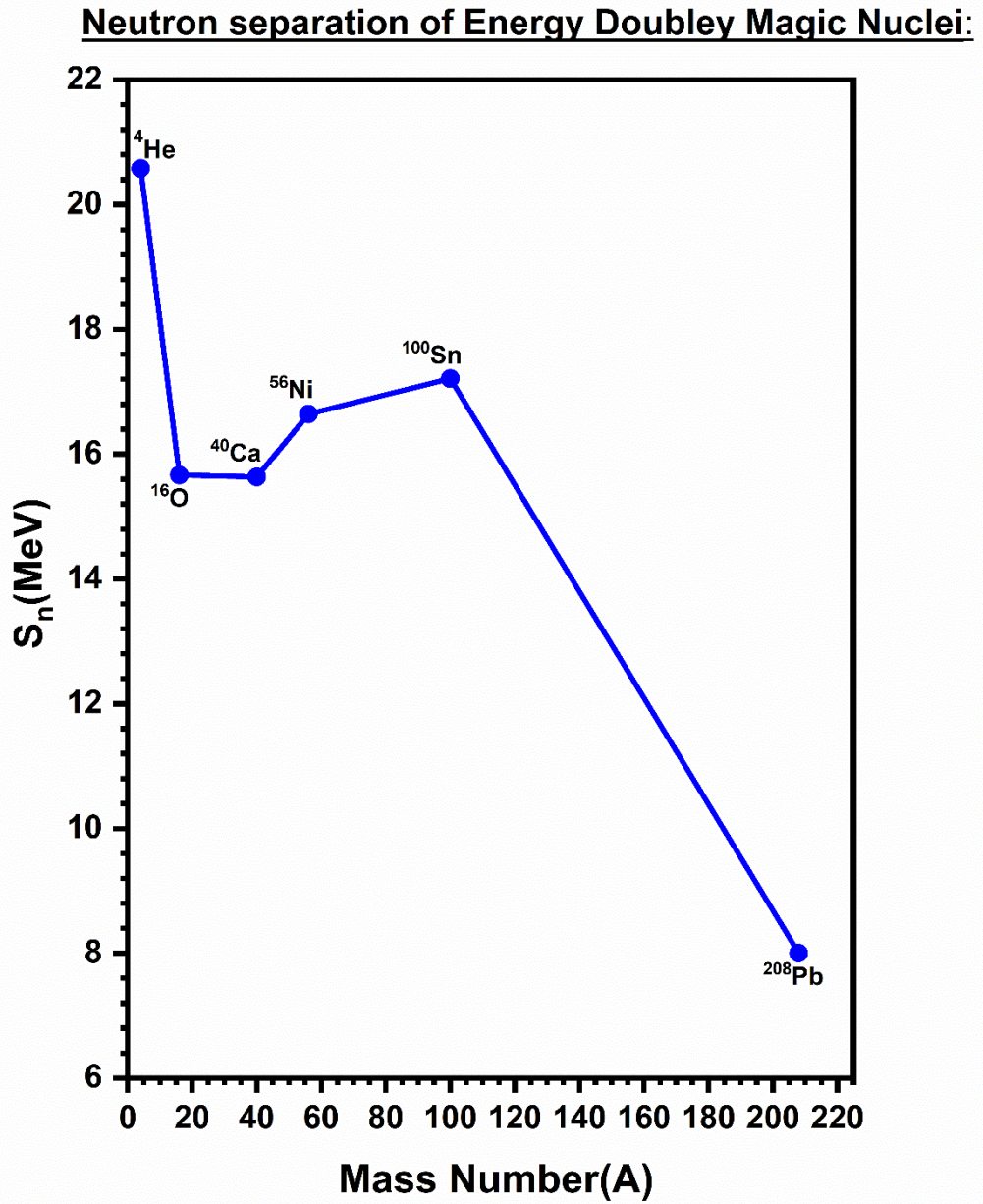
**Fig.3.12:** Proton Separation Energy of (Pb) isotopes plotted with Mass Number( $A$ )

**Table3.2:** Neutron and Proton Separation Energies for Doubly Magic Nuclei. [7]

| <b>Nuclei</b>                                     | <b><math>S_n(\text{MeV})</math></b> | <b><math>S_p(\text{Mev})</math></b> |
|---------------------------------------------------|-------------------------------------|-------------------------------------|
| <b><math>{}^4\text{He} (N,Z=2)</math></b>         | <b>20.57762</b>                     | <b>19.81387</b>                     |
| <b><math>{}^{16}\text{O}(N,Z=8)</math></b>        | <b>15.66396</b>                     | <b>12.12741</b>                     |
| <b><math>{}^{20}\text{Ca}(N,Z=20)</math></b>      | <b>15.66396</b>                     | <b>8.328159</b>                     |
| <b><math>{}^{56}\text{Ni}(N,Z=28)</math></b>      | <b>16.64297</b>                     | <b>7.166511</b>                     |
| <b><math>{}^{100}\text{Sn}(N,Z=50)</math></b>     | <b>17.211</b>                       | <b>3.0590</b>                       |
| <b><math>{}^{208}\text{Pb}(N=126,Z=82)</math></b> | <b>8.003</b>                        | <b>8.153369</b>                     |

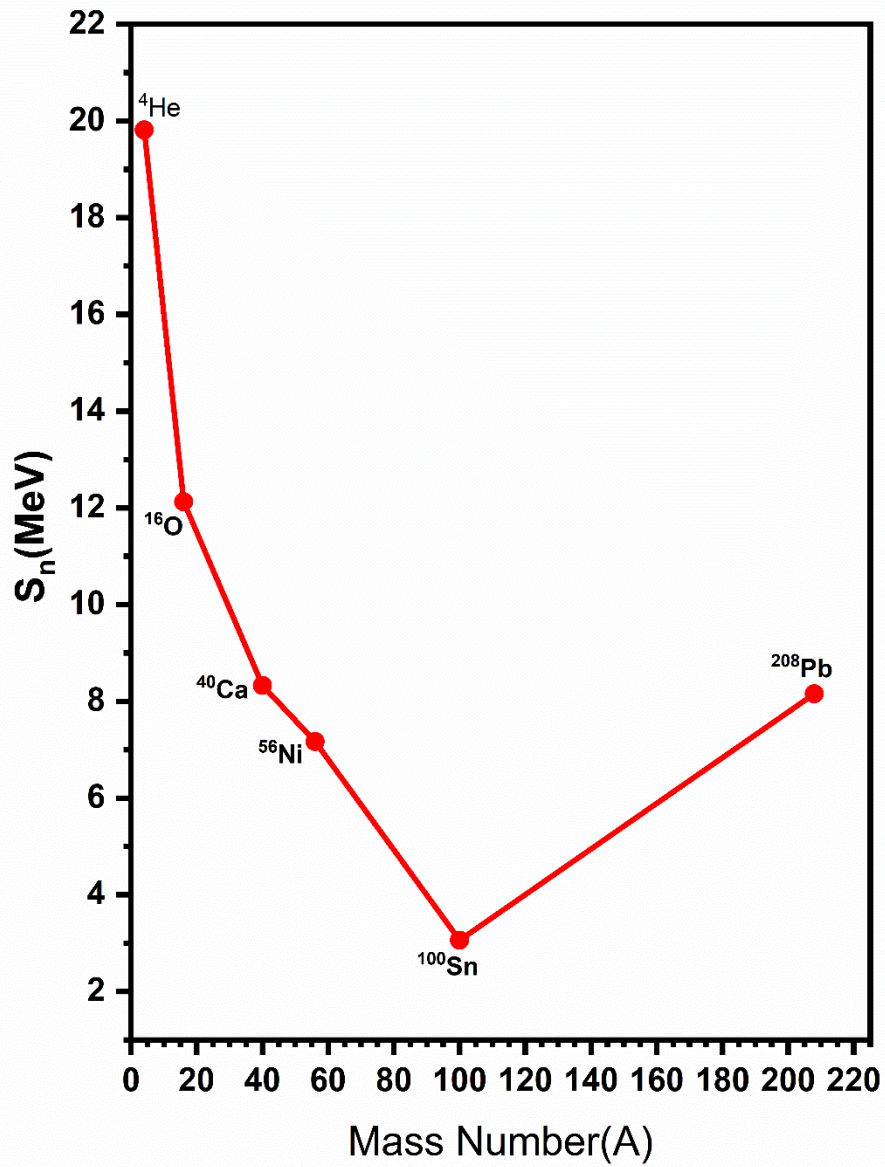
- Here calculated  $S_n$  and  $S_p$  for Doubly Magic Nuclei by mass and B.E like before Calculated for table 3.1

## $S_n$ and $S_p$ Graph For Doubly Magic Nuclei:



**Fig.3.13:** Neutron Separation Energy for all Doubly Magic nuclei( $^4\text{He}$ ,  $^{16}\text{O}$ ,  $^{40}\text{Ca}$ ,  $^{56}\text{Ni}$ ,  $^{100}\text{Sn}$ ,  $^{208}\text{Pb}$ ) Plotted with Mass Number( $A$ )

**Proton Separation Enrgy of Doblely Magic Nuclei:**



**Fig.3.14:** Proton Separation Energy for all Doubly Magic nuclei( ${}^4\text{He}$ ,  ${}^{16}\text{O}$ ,  ${}^{40}\text{Ca}$ ,  ${}^{56}\text{Ni}$ ,  ${}^{100}\text{Sn}$ ,  ${}^{208}\text{Pb}$ ) Plotted with Mass Number( $A$ )

### 3.2 Discussion: About Graphs

Here we made Graph of Separation Energy ( $S_n$  and  $S_p$ ) Vs. Mass Number

- In  $S_n$  The curves are zigzag due to Odd-Even Staggering. Nuclei with an even number of nucleons are more stable (higher energy point) than the nuclei with an odd number of nucleons next to them. (eg. Fig.3.11)
- The Proton Separation Energy ( $S_p$ ) tends to gradually increase as neutrons are added. This happens because adding neutrons increases the strong, attractive Nuclear Force without increasing the repulsive Coulomb Force (since the proton number,  $Z$ , stays the same), which effectively holds the protons more tightly. (eg. Fig.3.12)
- Here for Doubly Magic Nuclei We took Magic Proton( $Z=2,8,20,28,50,82$ ) and magic Neutron ( $N=2,8,20,28,50,82,126$ ) For these Nuclei ( ${}^4\text{He}$ ,  ${}^{16}\text{O}$ ,  ${}^{20}\text{Ca}$ ,  ${}^{56}\text{Ni}$ ,  ${}^{100}\text{Sn}$ ,  ${}^{208}\text{Pb}$ ) these are the Doubly Magic Nuclei for this we calculated Separation as we can see in table-3.2 after calculation we made graph all Magic Nuclei Separation Energy Vs Mass Number

## Chapter-4: Summary and Conclusion

### 4.1 Summary of Work:

The present work was focused on investigating the Neutron Separation Energy ( $S_n$ ) and Proton Separation Energy ( $S_p$ ) for isotopes of key magic nuclei (He, O, Ca, Ni, Sn, Pb). The primary objective was to demonstrate the existence of the Nuclear Shell Model and confirm the stability conferred by magic numbers ( $N, Z = 2, 8, 20, 28, 50, 82, 126$ ) through empirical calculation.

### 4.2 Conclusions and Results:

#### 4.2.1 Empirical Validation of Nuclear Shells:

The most significant finding is the observation of sharp discontinuities (sudden drops) in the separation energy curves immediately after a magic number is reached.

- Evidence of Shell Closure: The high value of separation energy before a magic number indicates that the shell is completely full and highly stable. For instance, the  $S_n$  required to remove a neutron from the closed-shell  $^{16}\text{O}$  is 15.66 MeV, but the energy required to remove the next neutron (which starts a new shell, in  $^{17}\text{O}$ ) drops dramatically to 4.14 MeV. This drastic drop provides direct evidence of the existence of large energy gaps between nuclear shells.
- Proof of Instability: The calculation for  $^5\text{He}$  (after the  $N=2$  magic number) yielded a Neutron Separation Energy of -0.74 MeV. A negative separation energy proves that this nucleus is unbound; it is so unstable that it will immediately decay by emitting a neutron, demonstrating the fragility of stability immediately following a shell closure.
- Doubly Magic Stability: Nuclei that are magic in both  $N$  and  $Z$  (like  $^4\text{He}$ ,  $^{40}\text{Ca}$ , and  $^{208}\text{Pb}$ ) exhibit the highest  $S_n$  and  $S_p$  values in their respective

regions, confirming their superior stability due to mutually reinforcing shell closures.

#### **4.2.2 The Odd-Even Staggering (Pairing Effect):**

The  $S_n$  curve exhibits a clear sawtooth pattern between magic numbers, where nuclei with an even number of neutrons are consistently more stable than their neighbors with an odd number of neutrons. This effect is a direct consequence of the pairing force in the nucleus.

- Higher  $S_n$  for Even-N: In even-N nuclei, the last neutron is paired with another neutron (with opposite spin), forming a strong, stable bond. To remove this neutron, extra energy (known as the pairing energy) is required to break the pair, resulting in a higher  $S_n$  value.
- Lower  $S_n$  for Odd-N: In odd-N nuclei, the last neutron is unpaired. Removing this single, unpaired nucleon does not require breaking a pair, resulting in a lower  $S_n$  value.

#### **4.2.3 The Increasing Trend in Proton Separation Energy ( $S_p$ ):**

For a chain of isotopes (constant  $Z$ , increasing  $N$ ), the  $S_p$  generally increases as the number of neutrons  $N$  increases. This trend is explained by the balance between the attractive Strong Nuclear Force and the repulsive Coulomb Force:

- Constant Repulsion: Since the number of protons ( $Z$ ) is constant across an isotope chain, the Coulomb repulsive force between the protons remains largely the same.
- Increasing Attraction: As more neutrons are added, the overall Strong Nuclear Force (attraction) acting on the protons increases significantly.
- Net Effect: The addition of neutrons strengthens the overall nuclear glue without increasing the electrical repulsion. Consequently, the protons are held more tightly, and more energy ( $S_p$ ) is required to remove a proton.

## References

1. Arthur Beiser (2003). Concepts of Modern Physics. 6th ed. (page No.380-412).
2. [https://www.citycollegekolkata.org/documents/online\\_course\\_materials/20200930\\_Binding\\_Energy.pdf](https://www.citycollegekolkata.org/documents/online_course_materials/20200930_Binding_Energy.pdf)
3. [https://sciencenotes.org/mass-defect-definition-and-formula/#google\\_vignette](https://sciencenotes.org/mass-defect-definition-and-formula/#google_vignette)
4. K. S. Krane (1987). Introductory Nuclear Physics. 1st Edition. John Wiley & Sons. ( 65–70).
5. The basics of Nuclear And Particle Physics(2010).  
[https://www.google.com/imgres?imgurl=https%3A%2F%2Fmedia.springernature.com%2Fflw685%2Fspringer-static%2Fimage%2Fchp%253A10.1007%252F978-3-030-80116-8\\_4%2FMediaObjects%2F500631\\_1\\_En\\_4\\_Fig5\\_HTML.png&tbnid=vgyTdIZqokMIUM&vet=1&imgrefurl=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-030-80116-8\\_4&docid=WQ1EFGgmKwlfNM&w=685&h=965&hl=en-IN&source=sh%2F%2Fim%2Fm1%2F4&kgs=667f747f023ff81d&shem=isst](https://www.google.com/imgres?imgurl=https%3A%2F%2Fmedia.springernature.com%2Fflw685%2Fspringer-static%2Fimage%2Fchp%253A10.1007%252F978-3-030-80116-8_4%2FMediaObjects%2F500631_1_En_4_Fig5_HTML.png&tbnid=vgyTdIZqokMIUM&vet=1&imgrefurl=https%3A%2F%2Flink.springer.com%2Fchapter%2F10.1007%2F978-3-030-80116-8_4&docid=WQ1EFGgmKwlfNM&w=685&h=965&hl=en-IN&source=sh%2F%2Fim%2Fm1%2F4&kgs=667f747f023ff81d&shem=isst)
6. K. Heyde (2020). Basic Ideas and Concepts in Nuclear Physics. IOP Publishing. (195-238).
7. <https://uen.pressbooks.pub/introductorychemistry/chapter/nuclear-fusion/>
8. R. B. Cakirli and R. F. Casten (2013). Nuclear binding and nuclear structure. International Journal of Mass Spectrometry, 349-350, (page No. 187–191).
9. The AME 2020 atomic mass evaluation (II). Meng Wang, W.J. Huang, F.G. Kondev, G. Audi, S. Naimi, Chinese Physics. C45, 03003 (2021)